



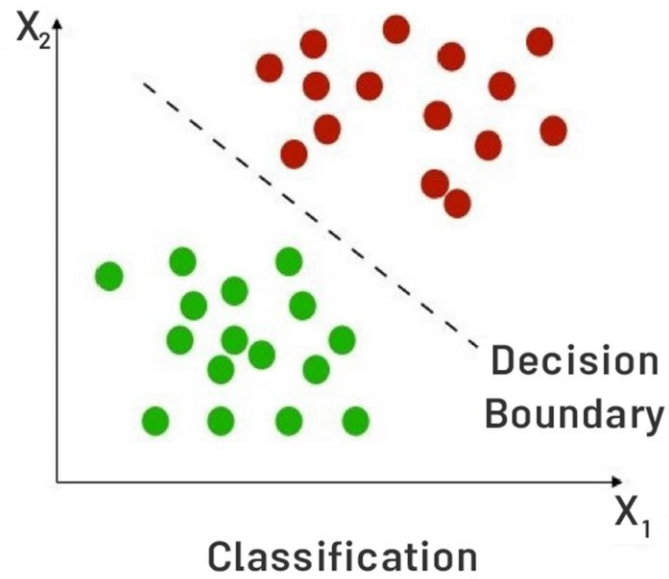
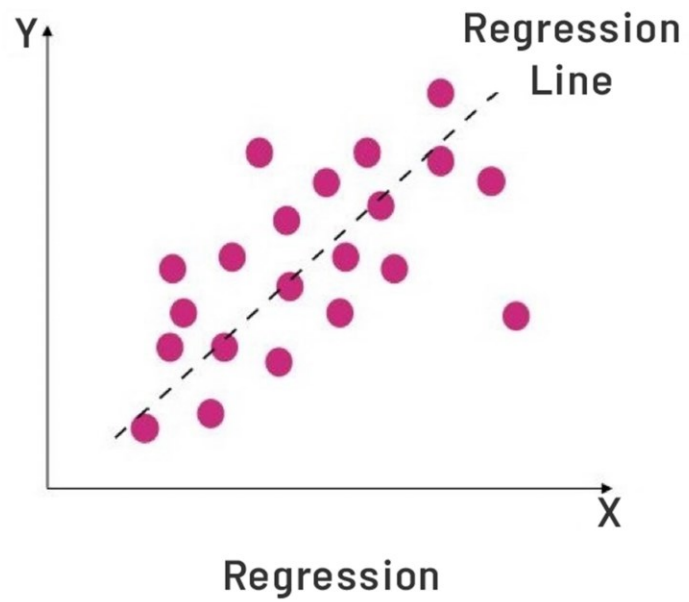
Classification: K-nearest neighbors

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Regression vs. Classification





K-nearest neighbors (K-NN)

► Binary classification:

$$h: X \rightarrow \{ +, - \}$$

► Old algorithm borrowed from pattern recognition

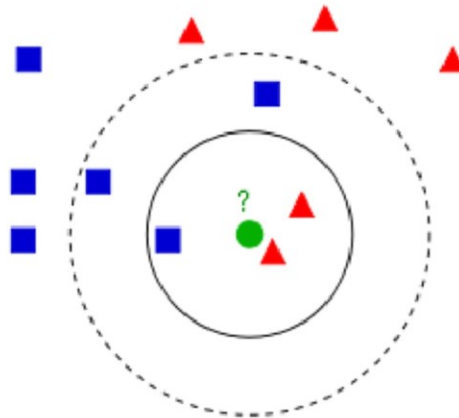
► K-NN Properties:

► Instance-based

► 1-NN, 3-NN, 5-NN, ...

► Lazy learning (vs. Eager)

► Not scalable





K-NN (cont.)

► K-NN Properties:

► Distance metrics

► Euclidean: $d(\mathbf{x}, \mathbf{x}') = \sqrt{\sum_i (x_i - x'_i)^2}$

► Weighted K-NN, where the i -th nearest neighbor is assigned a weight w_{ni} , with $\sum_{i=1}^n w_{ni} = 1$

► Manhattan, minkowski, Chebyshev, ...

► Non-parametric



K-NN (cont.)

► K-NN Properties:

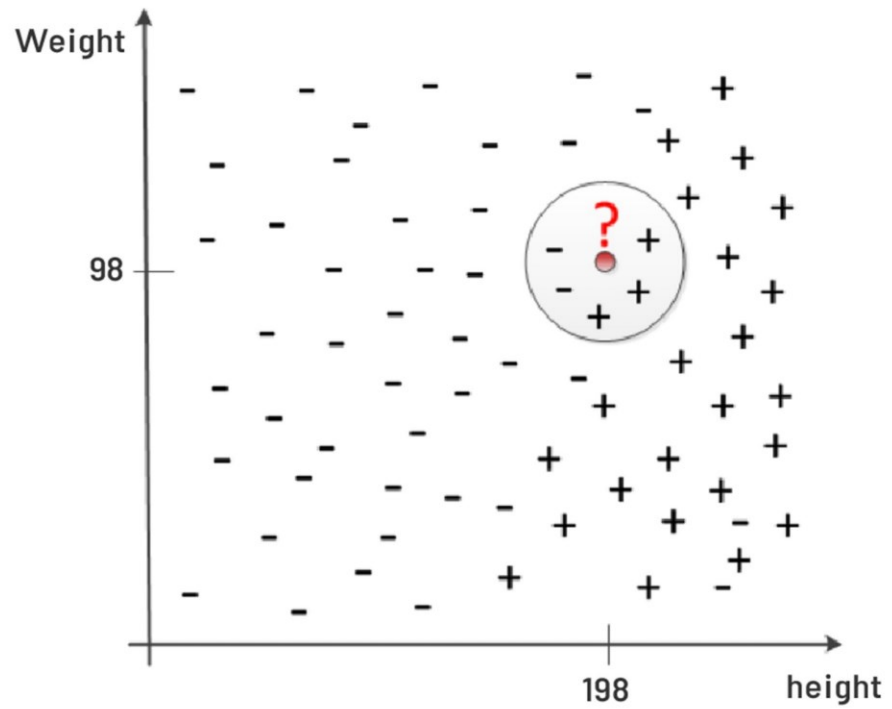
► Bias-Variance tradeoff





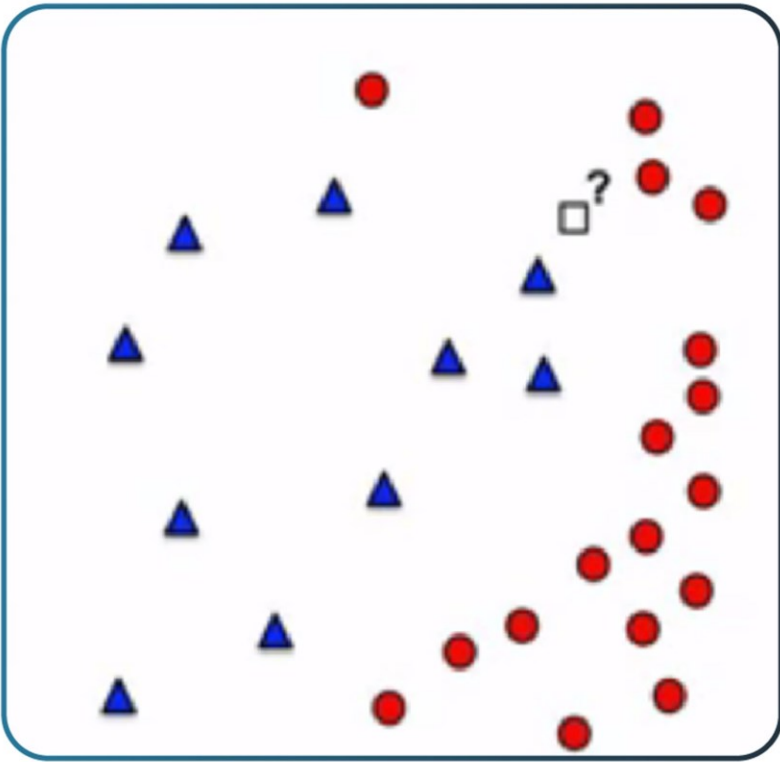
K-NN Example

► Who is good at Basketball?



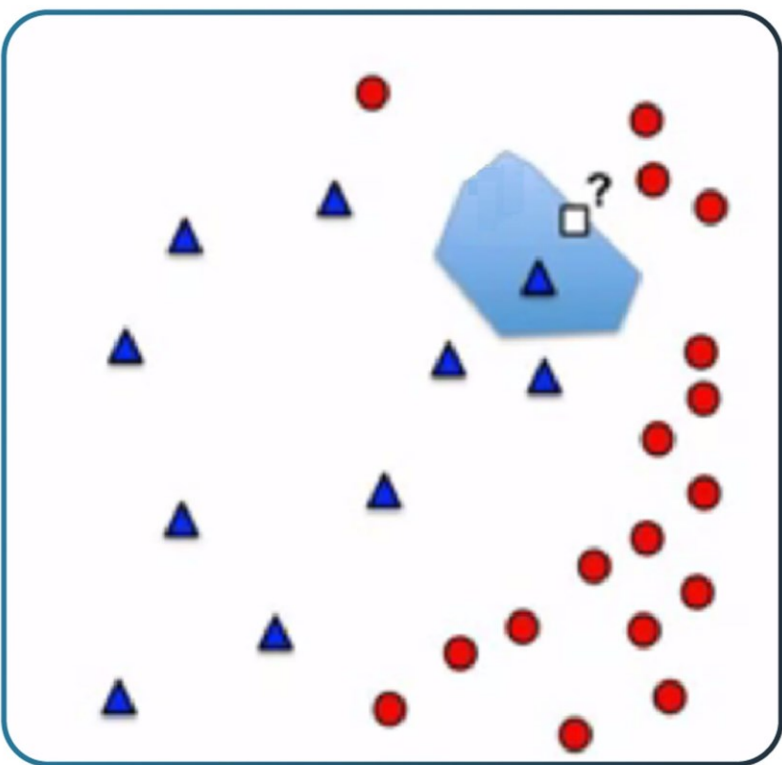


1-NN Decision Boundary





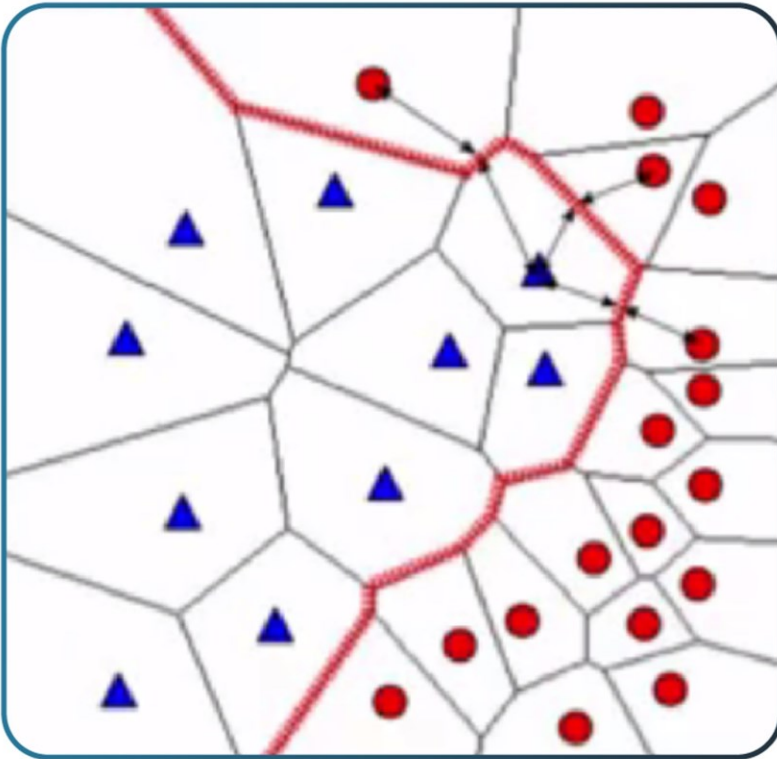
Voronoi Cell





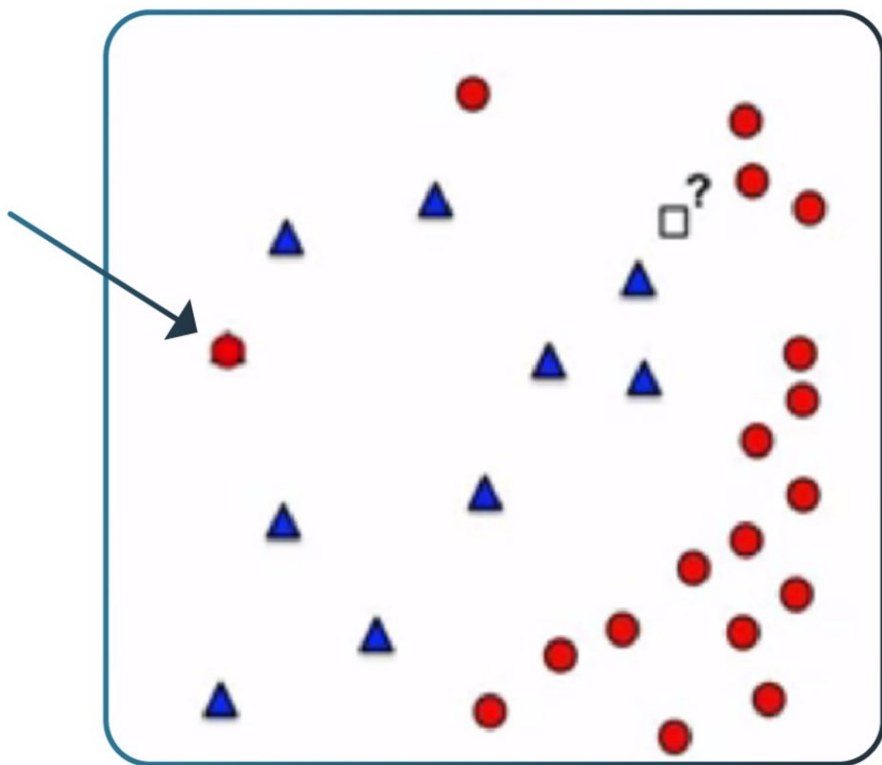
Voronoi Diagram

► Fortune's algorithm



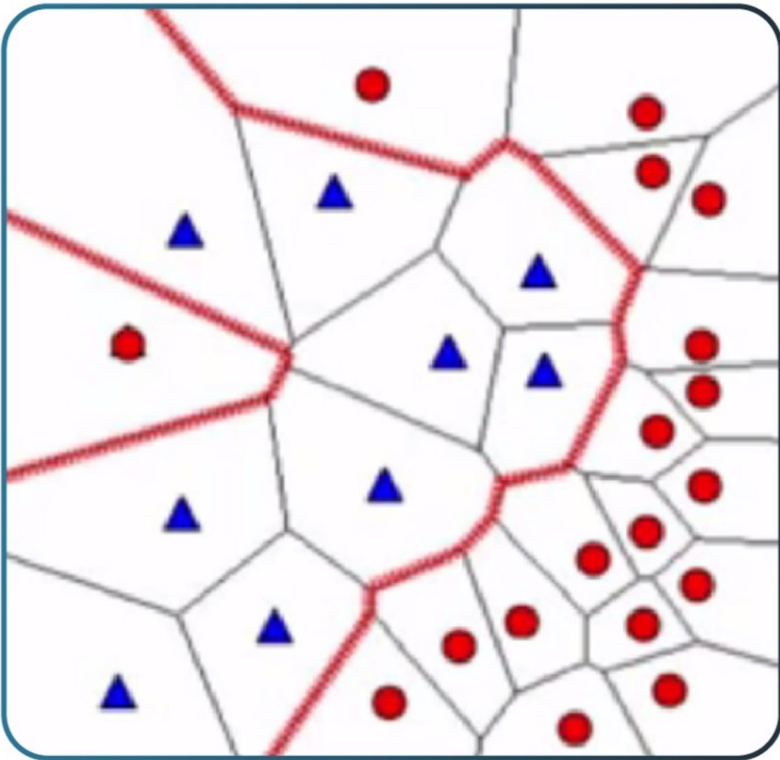


Noisy Example Added





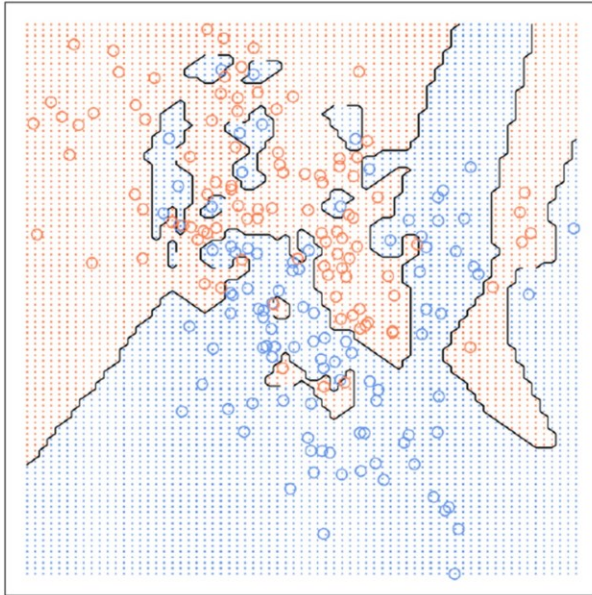
Sensitivity to Noise



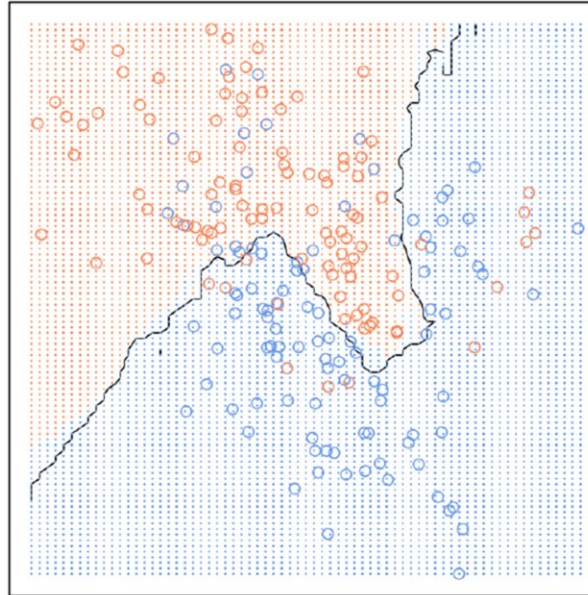


The higher the K, the smoother the Decision Boundary

1-nearest neighbours



20-nearest neighbours





Further Reading

- ▶ Weighted k-nearest neighbors
- ▶ K-NN for real valued prediction (regression)
- ▶ Curse of dimensionality in K-NN
- ▶ Sublinear time algorithm for approximating K-NN